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import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.metrics import mean_squared_error

# Step 1: Load data
data = pd.read_csv("house_data.csv") # Replace with your dataset

# Step 2: Select features and target
features = ['area', 'location', 'num_rooms', 'age'] # Adjust as needed
target = 'price'
X = data[features]
y = data[target]

# Step 3: Preprocessing
numeric_features = ['area', 'num_rooms', 'age']
categorical_features = ['location']

numeric_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='mean')),
    ('scaler', StandardScaler())
])

categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('encoder', OneHotEncoder(handle_unknown='ignore'))
])

preprocessor = ColumnTransformer(transformers=[
    ('num', numeric_transformer, numeric_features),
    ('cat', categorical_transformer, categorical_features)
])

# Step 4: Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Step 5: Choose model
model = GradientBoostingRegressor() # You can also try LinearRegression(), Ridge(), etc.

# Step 6: Create pipeline and train
pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('model', model)
])

pipeline.fit(X_train, y_train)

# Step 7: Predict and evaluate
y_pred = pipeline.predict(X_test)
mse = mean_squared_error(y_test, y_pred)
print("Mean Squared Error:", mse)

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